Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 08)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

0 1 Table 1 describes three genetically modified (GM) crops.

Table 1

GM crop	Gene source	Characteristic
Bt maize / cotton	Bacillus thuringiensis bacterium	Produces a toxin that kills insect pests
Roundup Ready soya	Agrobacterium	Resistant to the herbicide glyphosate
Golden rice	Daffodil and Erwinia bacterium	Produces vitamin A in the grain

0 1 . 1 Explain the main advantage of GM over conventional selective breeding.

[2 marks]

.....

.....

.....

.....

0 1 . 2 For each crop in Table 1, explain one potential benefit.

[3 marks]

Bt crops .....

.....

Roundup Ready soya .....

.....

Golden rice .....

.....

0 1 . 3 Explain **four** concerns about GM crops.

[4 marks]

.....

.....

.....

.....

.....

.....

Do not write outside the box

Do not write outside the box

.....

.....

.....

0 1 . 4 Give **one** reason why GM crops have been slow to enter cultivation in Europe.

[1 mark]

.....

.....

10

0 2 . 1 Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

[1 mark]

A Environmental DNA (eDNA) analysis ☐

B Radio tracking ☐

C Lincoln index (mark-release-recapture) ☐

D Leg rings ☐

0 2 . 2 Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

[2 marks]

.....

.....

.....

.....

0 2 . 3 Explain how DNA databases could help to control the illegal trade in elephant ivory.

[2 marks]

.....

.....

.....

.....

5

0 3 . 1 Explain **four** properties of water that make it essential for life.

[4 marks]

0 3 . 2 Describe how photosynthetic organisms changed the atmosphere and explain how this allowed life to colonise the land.

[4 marks]

0 3 . 3 Explain how carbon sequestration by early photoautotrophs helped maintain a suitable temperature on Earth.

[2 marks]

10

Do not write outside the box

0 4 . 1 Describe how agriculture alters the hydrological cycle.

[3 marks]

0 4 . 2 Explain how each of the following consumer choices can reduce the environmental impact of food production.

[4 marks]

Seasonal / local food

Organic food

0 4 . 3 Explain why post-war European governments introduced grants and guaranteed prices, and describe **one** environmental consequence.

[3 marks]

Do not write outside the box

10

0 5 A new pond was dug in a nature reserve. Table 1 shows the vegetation recorded at intervals over 60 years.

Table 1

Years after creation	Dominant vegetation	Water depth at centre / m
1	Single-celled algae, floating duckweed	2.0
5	Submerged pondweeds; reeds at margins	1.8
15	Reeds and reedmace over most of pond; water lilies	1.0
30	Marsh plants, sedges; willow and alder at edges	0.2
60	Willow / alder carr; oak seedlings	0 (soil waterlogged)

0 5 . 1 Name the type of sere shown in Table 1 and identify the pioneer species.

[2 marks]

.....

.....

.....

.....

0 5 . 2 Explain the changes shown in Table 1.

[5 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Do not write outside the box

Do not write outside the box

.....

0 5 . 3 Explain why a lithosere and a hydrosere in the same region end with the same climax community.

[2 marks]

.....

.....

.....

.....

0 5 . 4 Describe the early stages of a lithosere.

[1 mark]

.....

.....

10

0 6 . 1 Describe **three** methods of reducing seabird by-catch in long-line fisheries.

[3 marks]

1 .....

.....

2 .....

.....

3 .....

.....

0 6 . 2 Explain how restricting fishing effort and restricting fishing methods reduce overfishing.

[3 marks]

.....

.....

.....

.....

.....

.....

Do not write outside the box

0 6 . 3 Describe how captive rearing and release ('population seeding') can increase wild populations of lobsters or oysters.

[2 marks]

0 6 . 4 Explain why collecting data on planktonic fish eggs and larvae helps to prevent population collapse.

[2 marks]

10

0 7 A conservation organisation is converting 700 ha of arable farmland on the coast into a wetland reserve for breeding and wintering birds (as at Wallasea Island).

0 7 . 1 Explain why habitat area and shape are important in the design of the reserve.

[4 marks]

0 7 . 2 Explain how the reserve could be designed to provide a range of niches.

[3 marks]

Do not write outside the box

0 7 . 3 Explain why some species may need to be deliberately introduced while others colonise naturally.

[2 marks]

0 7 . 4 Explain the value of a biological corridor linking the reserve to other wetlands.

[1 mark]

10

0 8 A student sampled the ground-flora of two woodlands. Table 1 shows the results for woodland A.

Table 1

Species	Number of individuals (n)
Bluebell	40
Wood anemone	30
Dog's mercury	20
Bramble	10
Total (N)	100

0 8 . 1 Use the formula $D = N(N - 1) / \Sigma n(n - 1)$ to calculate Simpson's index of diversity for woodland A.

[3 marks]

Do not write outside the box

Answer

0 8 . 2 Woodland B had an index of 1.6. State what this shows and give **one** limitation of using the index alone to compare the woodlands.

[2 marks]

.....

.....

.....

.....

5

0 9 . 1 Explain how natural material cycles differ from linear human systems of material use.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

0 9 . 2 Explain why the use of toxic metals in electronic products is a particular problem for sustainability.

[2 marks]

Do not write outside the box

0 9 . 3 Describe how the design of a product can move it from a linear to a cyclical system.

[4 marks]

10

1 0 . 1 Explain how the relationship between coral polyps and zooxanthellae allows a high biomass of life in nutrient-poor tropical seas.

[4 marks]

1 0 . 2 Explain why growing coral reefs help to resist climate change.

[2 marks]

1 0 . 3 Discuss the importance of tropical coral reefs and evaluate the methods used to conserve them.

[9 marks]

Do not write outside the box

Do not write outside the box

Extra space

15

Write an essay on one of the following topics.

1 1 . 1 Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

OR

1 1 . 2 Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

END OF QUESTIONS